

Invigilation System: Operations & User Manual

A Step-by-Step Operator Guide for Academic Administration Staff

Document Class: Software Operations & Training Guide

Prerequisites: No programming skill required (Point-and-Click interface)

Deployment Server: <http://localhost:8080>

1. Starting up the Scheduler Website

Step 1: Open the Application Folder

- Open the folder named **IIT Bombay** on your computer's Desktop.
- Locate the file named **start_invigilation_server.bat** (identifiable by a cog/gear icon or Windows Batch File type).

Step 2: Start and Access

- **Double-click the file:** A black window will briefly pop up and minimize.
- **Web Browser Opens:** Your default browser will load the portal automatically at:
http://localhost:8080
- **Automated Roster:** All 46 faculty members are loaded automatically from the Excel template sheet.

2. Customizing Faculty & Exam Sessions

Faculty Roster Customization • Availability Checks: Select a teacher and uncheck specific session dates to mark leaves or out-of-station blocks. • Lecture Conflict Blocks: Check slots under PG Timetable Blocks to ensure teachers are not scheduled during their classes. • Autosave Sync: All checkboxes are saved to the storage disk instantly.	Exam Sessions Setup • Exams Configuration: Create new exam dates, select Morning/Afternoon sessions, input course codes, and assign rooms. • Staff Capacity: Specify the exact number of required invigilator personnel needed for each room.
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3. Generating Duties & Simple Troubleshooting

Generating the Schedule Grid

- **One-Click Solve:** Click the blue  **Run Allocation** button at the top right.
- **Fair Distribution:** The engine runs optimization algorithms to allocate duties proportionally, satisfying all constraints in under 2 seconds.
- **Verify Fairness:** Review Jain's Index dial (targets close to 1.00) to confirm equitable load balance.

Troubleshooting Errors

- **Unfeasible Schedule Warning:** If a red warning appears saying '*Solver failed to generate a feasible allocation*', too many teachers are blocked from duty.
- **Resolution:** Go back to the Faculty Roster tab, unblock leaves or suspend PG lecture clashes for highly restricted faculty, then re-run allocation.